

Friction of Ionic Liquid–Glycol Ether Mixtures at Titanium Interfaces: Negative Load Dependence

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The atomic force microscopy experiments and nonequilibrium molecular dynamics (NEMD) simulations demonstrate a negative friction–load dependence to ionic liquid–glycol ether mixtures, that is, the friction decreases as the normal load increases. NEMD simulations reveal a structural reorientation of the studied ionic liquid (IL): as the normal load increases, the cation alkyl chains of ILs change the orientation to preferentially parallel to the tip scanning path. The flat-oriented IL structures, similar to the “blooming lotus leaf,” produce a new sliding interface and reduce the friction. A further molecular dynamics simulation is carried out by adopting slit-pore models to mimic the tip approaching process to confirm the dynamics of ILs. A faster diffusion of ILs in the smaller slit pore is observed. The faster diffusion of ILs in the more confined slit pore facilitates the structural reorientation of ILs. The resulted new sliding surface is responsible for the observed smaller friction at higher loads, also known as the negative friction–load dependence. These findings provide a fundamental explanation to the role of ILs in interfacial lubrications. They help to understand liquid flow properties under confinement, with implications for the development of better nanofluidic devices.

1. Introduction

Friction, as one of the oldest phenomena, is one of the major causes that result in energy and material losses in mechanical processes,^[1] and plays an important role in the history of mankind and in particular of natural science. The milestone of friction mechanism was established in the 17th century^[2–4] that friction force is linearly proportional to the normal load. This theorem, also known as Amontons’ law, in which the monotonic increase of friction with load has been observed across a range of length scales.^[5] Amontons’ law describes the friction behavior in the majority of rubbing surfaces, involving both dry and lubricated contacts, both rough and smooth surfaces, both ductile and brittle, and both macroscopic and microscopic contacts.^[5]

Despite the successfulness of Amontons’ law at macro- and microscales, one has to pay special attention to friction-relevant quantities at micro- and nanolevels. Especially, the frictional phenomena are

more complex, when it comes to soft contact interfaces, such as those composed of liquid/liquid or liquid/solid interfaces, because interfacial structural transitions at higher normal load might occur. For example, recent friction studies of lamellar graphite surface revealed a negative friction coefficient during the atomic force microscopy (AFM) tip retraction process, i.e., the friction force increases as the normal load decreases.^[6] Such negative correlation between friction force and normal load is not inherent in the Amontons’ law. Yet, it has been observed from various systems (e.g., graphite, graphene, polymers, or other 2D materials,) and under different conditions (such as vacuum).^[7–12]

In the graphene system, the negative friction coefficient is attributed to the partial delamination (exfoliation) of topmost graphene layers. When the AFM tip retracts, the tip–surface adhesion force can overcome graphite exfoliation energy. Thus, the tip will lift the upper graphene layer(s), and separate them locally from the bulk. Therefore, during the friction force measurement, a ripple is formed around the AFM tip. In addition, both the ripple height and width will change dynamically in response to the tip retraction and the resulted exfoliation of topmost graphene layers. Such nanoscale dynamic change of

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the sliding interface leads to a fundamentally different friction coefficient, which, on the other hand, is not incorporated in the Amontons' law. This lack explains why both positive and negative friction-load dependence were observed on the same suspended graphene in different load regimes, from AFM experimental^[7] and atomistic simulation^[8] studies.

In this work, we report another type of negative friction-load dependence by ionic liquid (IL)/glycol ether mixtures using AFM experiments and molecular dynamics simulations. The phosphonium-based IL/glycol ether mixture was applied onto titanium substrate as lubricant. We employed IL/oil mixtures in our experiments, because of the high viscosity of neat ILs, which would result in a high friction force.^[13] More importantly, the main cause of friction in hydrodynamic and elastohydrodynamic regimes is hydrodynamic drag, which consumes 10–15% of the total energy and is proportional to the lubricant viscosity.^[14] Also, the neat ILs as lubricants are more expensive compared with conventional base oils.^[15–17] One method to improve ILs lubricating efficiency and reduce the cost is to use less viscous lubricants by employing ILs as additives in base oils,^[9] and we therefore investigate in this study the friction behavior of IL/oil mixtures at Ti interfaces.

Nonequilibrium molecular dynamics (NEMD) simulations in our work revealed a similar negative load-dependent friction behavior as observed in our AFM experiment, and exhibited the IL structural reorientation during tip sliding. When the normal load increases, the cation alkyl chains change their orientation, generating a new interface for tip sliding with smaller friction force. This structural transition was further supported by calculating self-diffusion properties of the studied IL. With a slit-pore model, mimicking the confinement environment near the tip, we observed a faster self-diffusion of IL, implying that the IL will diffuse faster to facilitate the structural reorientation and the new sliding interface formation at larger normal loads.

2. Results and Discussion

We studied a combination of ionic liquid anions and cations, and their mixture with the base oil, diethylene glycol dibutyl ether (DEGDBE). The chemical structures of the phosphonium cations, $[P_{4,4,4,8}]^+$, $[P_{6,6,6,14}]^+$ and anions, $[BOB]^-$, $[BMB]^-$, $[DCA]^-$, $[BSCB]^-$, as well as the base oil DEGDBE are shown in Figure 1.

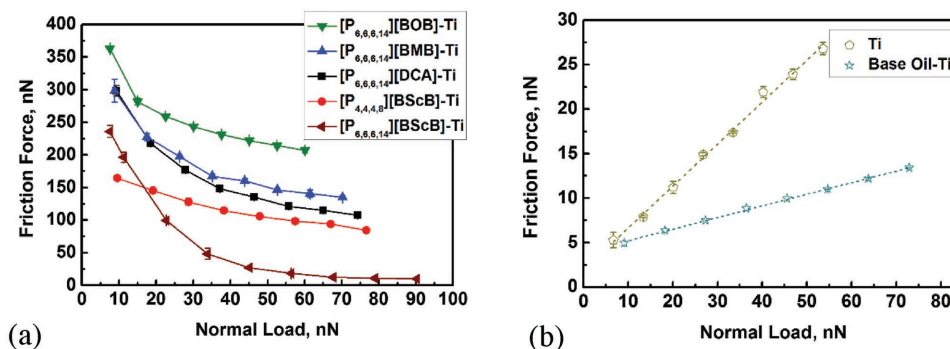


Figure 2. Friction measurements for a) the IL-base oil mixtures coated Ti interfaces, where 75 wt% of ILs, namely, $[P_{6,6,6,14}][BOB]$, $[P_{6,6,6,14}][BMB]$, $[P_{6,6,6,14}][DCA]$, $[P_{4,4,4,8}][BSCB]$, and $[P_{6,6,6,14}][BSCB]$, have been studied; b) bare Ti and base oil lubricated Ti interfaces.

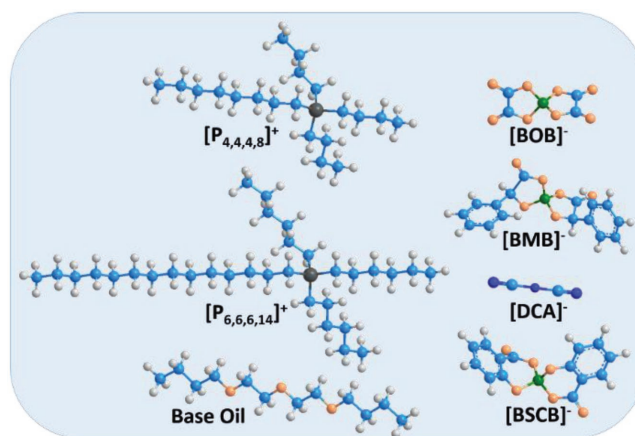


Figure 1. Structures of two cations, $[P_{4,4,4,8}]^+$ and $[P_{6,6,6,14}]^+$; four anions, $[BOB]^-$, $[BMB]^-$, $[DCA]^-$, and $[BSCB]^-$; base oil DEGDBE. Color codes are: P (black), B (olive), C (blue), H (gray), O (orange), N (navy).

The 75 wt% IL/oil mixtures in our work demonstrate the similar viscosity dependence on anions of ILs (Table S1, Supporting Information) as neat ILs.^[18] The dependence of densities for IL/oil mixtures on anions also follows the same trend observed in neat ILs.^[18] The viscosity has significantly decreased upon the addition of DEGDBE to the IL. This is indicative of the addition of base oil has no influence on the dependence of viscosities and densities on anions observed in our IL/oil mixtures. Our experience has shown that surface wetting properties exhibit strong effects on the interfacial friction.^[19] We thus examined the IL/oil wettability on Ti substrates, and observed no significant differences in contact angles (Table S1, Supporting Information) of different IL/oil mixtures on Ti substrates, indicating the wetting behavior for different IL/oil mixtures on Ti surfaces is similar. So we can exclude the possibility that the following different friction forces of different IL/oil systems result from the interfacial wetting behavior.

2.1. Negative Friction–Load Dependence in IL/Oil Mixtures at Titanium Interfaces Observed by AFM

The friction properties of the IL/oil mixture coated Ti surfaces were probed by AFM with the 20 nm radius Si_3N_4 tip. As shown in Figure 2a, a negative friction-load dependence was observed

for all the studied 75 wt% IL/oil mixtures at Ti interfaces, that is, the friction force decreases as the normal load increases. This negative load-dependent behavior was not only observed in 75 wt% IL mixtures at Ti interfaces, but also in the neat IL-Ti system (Figure S2a, Supporting Information) and 25 wt% IL/oil (1/3 in wt/wt) mixtures at Ti interfaces (Figure S2b, Supporting Information). In this work, we focused on the 75 wt% IL systems to reveal the cause of the observed phenomenon. A systematical study is undergoing to investigate substrate effect, and whether the negative friction-load dependence can be extended to other IL systems. It is worth noting, as shown in Figure 2a, anomalously larger friction forces than normal load were observed in 75 wt% IL systems, however, this phenomenon disappeared in 25 wt% IL systems (Figure S2b, Supporting Information), and occurred again in the neat IL $[P_{6,6,6,14}][BOB]$ coated Ti substrates (Figure S2a, Supporting Information). It is unusual to observe larger friction forces than normal load, but it happens, in different ways. For example, large friction forces were observed for a metal-coated AFM tip on mica in UHV at high loads, due to the wear of the surface.^[20] In the case of an IL between negatively charged surfaces, the stable IL film structures contain different layers of ions. When the interlocking of those ion layers is strongly ordered, the resulted friction force could also be larger than the normal load.^[21,22]

We attribute larger friction forces observed in high IL concentration systems (75 and 100 wt%) to the high viscosity. Because the main cause of friction in hydrodynamic/elastohydrodynamic regimes is hydrodynamic drag, which is proportional to the lubricant viscosity.^[14] This indicates that highly viscous ILs could result in a larger friction force.^[13] As we expected, viscosities in 75 wt% IL/oil mixtures are much higher than that in 25 wt% mixtures (Tables S1 and S2, Supporting Information), and friction forces in latter systems are smaller than normal load. So a small amount of the addition of ILs in base oil is favorable in reducing friction, and it is necessary for us in a near future to explore the friction behavior in IL/oil mixtures by adding small amounts of ILs in the base oil.

In Figure 2a, the lowest friction is observed at the Ti interfaces coated with IL containing $[BScB]^-$ anion. By comparing the two ILs containing $[BScB]^-$ anion, the IL containing phosphonium cation $[P_{6,6,6,14}]^+$ with longer chain provided lower friction forces. On the contrary, without ILs, for the bare Ti surface and the Ti surface lubricated by the neat base oil (DEGDBE), there is a positive friction-load dependence: the measured friction force increases linearly when the normal load increases, following existing tribological models previously reported.^[9,15,23] The fitted friction coefficients from Figure 2b were 0.48 and 0.13 for the bare and the base oil lubricated Ti surfaces, respectively.

Pertinent questions are why “negative” situation of friction-load dependence occurs at titanium surfaces lubricated with 75 wt% IL/oil mixtures, but not at the bare Ti surface and the Ti surface lubricated with neat DEGDBE. What is the origin of the negative friction-load dependence for IL/oil mixtures at Ti interfaces? The resolution of the friction data is insufficient to determine the observed “negative notion,” friction loops and molecular simulation would be performed further to manifest the friction-load relationship.

2.2. Interfacial Chemical and Topographical Properties

Friction properties can be greatly influenced by chemical composition changes of the interface. The Ti surfaces coated by the $[P_{4,4,4,8}][BScB]$ -base oil mixture are examined by X-ray photoelectron spectroscopy (XPS) before and after friction tests in order to get deeper insights into any chemical reactions occurring at the interfaces. Figure 3a shows the survey spectra of the $[P_{4,4,4,8}][BScB]$ -base oil coated Ti surface. It clearly reveals the presence of C 1s, O 1s, P 2p, B 1s on the surface, before and after AFM measurements, and no significant changes were observed in binding energies (<0.1 eV). This indicates the surface compositions of IL/oil mixtures at Ti interfaces are not influenced by AFM sliding procedure. In addition, no new peaks were observed in the high-resolution spectrum of the C 1s after AFM friction measurements (Figure 3b). As seen in the high-resolution spectrum of the C 1s electron region before and after AFM measurements, three major carbon-containing functional groups are present at 284.7, 286.2, and 288.6 eV, respectively. The C 1s peak component at 284.7 eV ($C-C/C=C$)^[24] is attributed to alkyl chains of the $[P_{4,4,4,8}]$ cation and DEGDBE, as well as the aromatic ring of the $[BScB]$ anion. A distinct C–O peak at 286.2 eV and the presence of $C(=O)-O$ bond at 288.6 eV^[24] are assigned to the aromatic ring of the $[BScB]$ anion. Moreover, electron regions of O 1s, P 2p, B 1s + P 2s (Figure S3, Supporting Information) exhibit in the same way before and after AFM sliding procedure. The XPS analysis was also performed for other systems involving $[P_{6,6,6,14}][BOB]$ -, $[P_{6,6,6,14}][BMB]$ -, $[P_{6,6,6,14}][DCA]$ -, $[P_{6,6,6,14}][BScB]$ -base oil mixtures at Ti surfaces. Comparing the spectroscopy information before and after the AFM measurements, no changes in both the shape of signal and binding energy values were observed, demonstrating no chemical reactions occurred during AFM sliding, see Figures S4–S7 of the Supporting Information.

For the Ti surface coated by the $[P_{4,4,4,8}][BScB]$ -base oil mixture, two typical examples were presented in Figure 4: one is the topographic image at a small normal load, 9.6 nN, and the RMS roughness is 97.9 nm; the other is at a large normal load, 76.6 nN, and the RMS roughness is 98.5 nm. It is apparent that no significant disruption (or roughness change) was observed as the load increased. Similarly, this absence of disruptions holds true to other studied systems, as shown in Figure S8 of the Supporting Information.

2.3. Friction Loops Dependence

As illustrated in Figure 5, at smaller normal loads, e.g., $F_N = 28.7$ nN, AFM tip shears against the IL/oil mixture layer far away from the Ti surface. However, the probe penetrates the near surface layer^[25–27] at higher loads, e.g., $F_N = 76.6$ nN, and shears against a different IL/oil mixture layer which can be substantially suppressed on the Ti surface. From the recorded high-resolution traces, where the friction force is plotted as a function of the lateral displacement, the friction does not always remain as a constant and changes locally. As illustrated in Figure 5, a decrease of the friction loop width was observed when the normal load was increased from 28.7 to 76.6 nN. The sliding dynamics analysis of other systems are listed in Figure S9

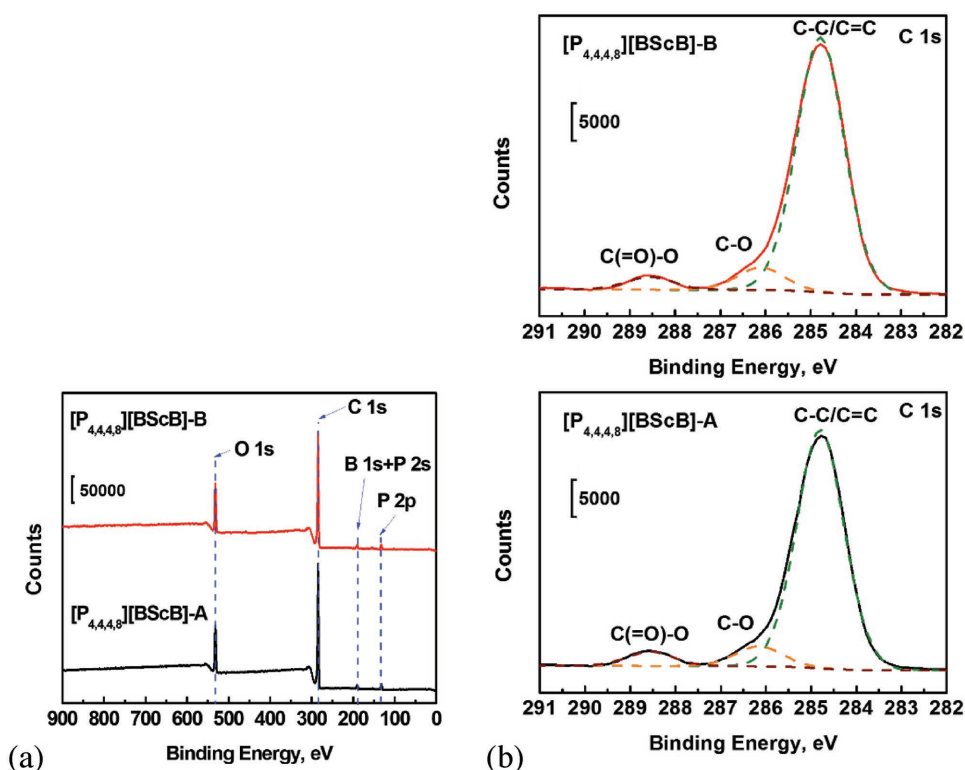


Figure 3. XPS spectra of $[P_{4,4,4,8}][BScB]$ -oil mixtures at Ti interfaces, before ($[P_{4,4,4,8}][BScB]$ -B) and after ($[P_{4,4,4,8}][BScB]$ -A) AFM friction measurements: a) survey, b) high resolution C 1s scans.

of the Supporting Information. It is concluded that the frictional interaction strength of the studied IL/oil mixtures, which is the difference between the forward and the backward friction traces, demonstrated the order: $[P_{6,6,6,14}][BOB] > [P_{6,6,6,14}][BMB] > [P_{6,6,6,14}][DCA] > [P_{4,4,4,8}][BScB] > [P_{6,6,6,14}][BScB]$. Such order is in agreement with the friction force measurements of Figure 2a. The frictional interaction strength corresponds to the magnitude of dissipation energy due to friction, and provides a quantitative measurement of the energy lost during a friction loop.^[28]

2.4. Negative “Friction–Load Dependence” Observed by Simulation

In order to reveal the fundamental mechanism at the molecular level, we performed a series of NEMD simulations to study the

atomic-scale sliding friction in ILs that is not accessible by AFM. Since the negative friction–load dependence was observed only when the ILs were used in the AFM experiments. We observed also a negative load-dependent behavior in the neat IL–Ti system (Figure S2a, Supporting Information), so we adopted the neat IL $[P_{4,4,4,8}][BScB]$ as the model to simplify the friction calculation process. The AFM tip was modeled as a capped carbon nanotube. **Figure 6** shows the calculated friction force as a function of different normal loads, and a smaller friction was observed when increasing the normal load. The simulation captures the same trend of negative frictional load dependence, although there are differences in the magnitude of the friction and the load. These discrepancies may be associated with the different size, material, and shape of the tips in the experiment and simulation.^[8] Similar to the AFM experimental findings, the larger friction forces than normal load were also observed

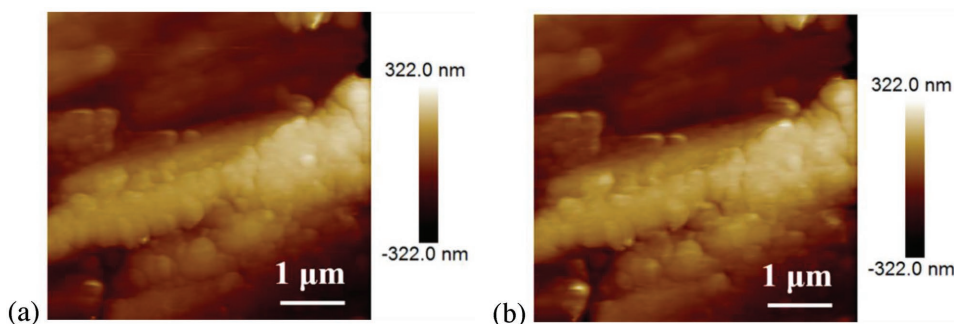


Figure 4. Topographic images of $[P_{4,4,4,8}][BScB]$ -oil mixtures at Ti interfaces at a) smaller normal load of 9.6 nN and b) larger normal load of 76.6 nN.

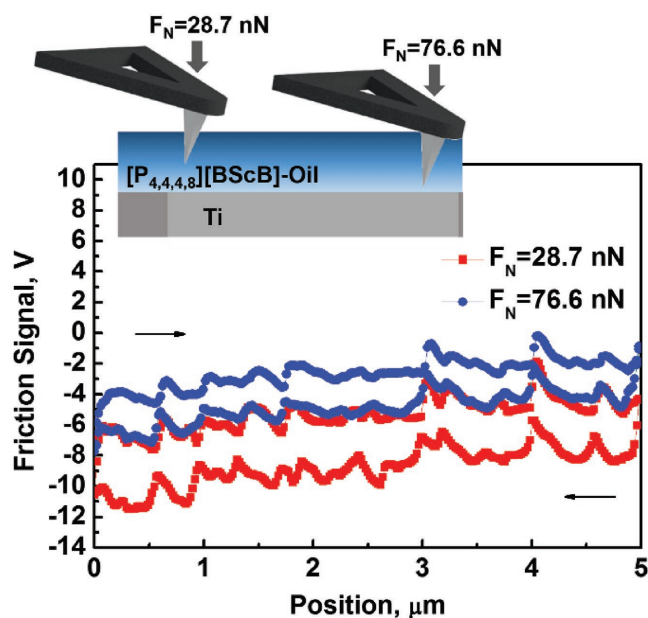


Figure 5. Typical friction loops taken for $[P_{4,4,4,8}][BScB]$ with concentration of 75 wt% in oil at Ti interfaces at two different normal loads (F_N). Friction values, here given in V, are calculated as the average difference for each interfaces, in the friction loop between the back and forth scans. The back-and-forth displacement is in the directions indicated by black arrows.

in the neat IL system using NEMD simulations, and we attributed this unusual phenomena still to the high viscosity. The similar friction trends observed in NEMD simulations to that in AFM experiments are indicative of the rationality of our simulations. The fact that both experiments and simulations exhibit the same trending, despite their very different speeds, tips, and substrates, suggests the mechanisms observed in simulations are indeed applicable to the corresponding experiments.

2.5. Structural Changes of Ionic Liquids

The changes in ion packing, i.e., structural changes, as the AFM tip penetrates the ILs layer at higher loads, were further

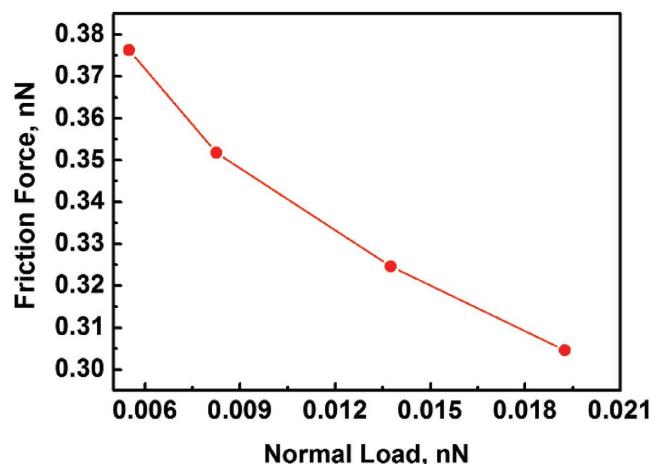


Figure 6. Friction as a function of normal load in the $[P_{4,4,4,8}][BScB]$ case, from NEMD simulations.

observed in the $[P_{4,4,4,8}][BScB]$ system around the tip during AFM scanning. The orientational probability of cations and anions was an excellent way to interpret the local structure of the ionic liquid. To determine the orientation behavior of cations and anions at the tip scanning interface, we defined the vector $P_{4,1}$, $P_{4,2}$, $P_{4,3}$, P_8 , and B_1 , B_2 of cation and anion respectively using molecular simulation as shown in insets in right panels of Figure 7a,b. We then defined the angle θ of the alkyl chain vector $P_{4,1}$, $P_{4,2}$, $P_{4,3}$, P_8 in the cation $[P_{4,4,4,8}]^+$ with respect to the interface normal vector z , and the angle ψ of the vector B_1 , B_2 in the anion $[BScB]^-$ to the normal vector z .

We evaluated the probability distribution of θ and ψ for the cation alkyl chains $P_{4,1}$, $P_{4,2}$, $P_{4,3}$, P_8 , and the anion B_1 , B_2 around the AFM tip during scanning. In the case of the cation shown in Figure 7a, the θ distribution for alkyl chains is dominated by large peaks around $\theta \approx 60^\circ$, 90° , 130° , 90° (orientation I), $\theta \approx 150^\circ$, 90° , 130° , 90° (orientation II), which indicates that two of the cation alkyl chains are orientated perpendicular to the normal vector z , with two other chains tilted pointing to z (see the sketch map in Figure 8a). The cation alkyl chains around the tip preferentially orient parallel with the tip scanning route, as the normal load increases. The resulted flat-oriented cation alkyl chains at a higher normal load, would be favorable in reducing friction force, and be responsible for the “negative friction–load dependence.” For the anion, two main preferential orientations with tilts exist, i.e., orientation I with peaks at $\psi \approx 50^\circ$, 110° , orientation II with peaks at $\psi \approx 110^\circ$, 135° , corresponding to the dominant orientation of the anions already seen in the sketch map in Figure 8b. Specifically, in both orientations I and II of anions, one of the benzene rings points to the scanning tip, leading to an electrorepulsion because of delocalized π electrons in benzene ring and tip, which may contribute to the lower friction. Nevertheless, the distributions in both cations (right in Figure 7a) and anions (right in Figure 7b) in the bulk system show almost no orientation ordering, i.e., no preferred orientation, in comparison with the orientation ordering in the near-tip layer.

2.6. Diffusion Coefficient-Structural Changes

Such structural reorientations observed in Figures 7 and 8 are possible if the diffusion is enhanced for the IL molecules. To support the enhanced diffusion hypothesis for the parallel oriented structure changes of ILs as the AFM tip penetrates the ILs layer at higher loads, we designed three slit-pore models to confine the neat $[P_{4,4,4,8}][BScB]$ using molecular dynamics simulation. The NEMD and the diffusion calculations are connected but not identical, and the diffusion calculations have qualitatively verified the enhanced diffusion mechanism. Our goal is to compare IL diffusion between the tip and the substrate during the AFM tip approaching to help us understand structural changes of ILs. In AFM experiments and NEMD friction calculations, the tip moves closer to the substrate as the external normal load increases. So we employed slit-pore models to mimic the tip approaching the substrate by varying the confinement width.

It may be expected that confined ILs exhibit slower self-diffusion, such as the confined octamethylcyclotetrasiloxane

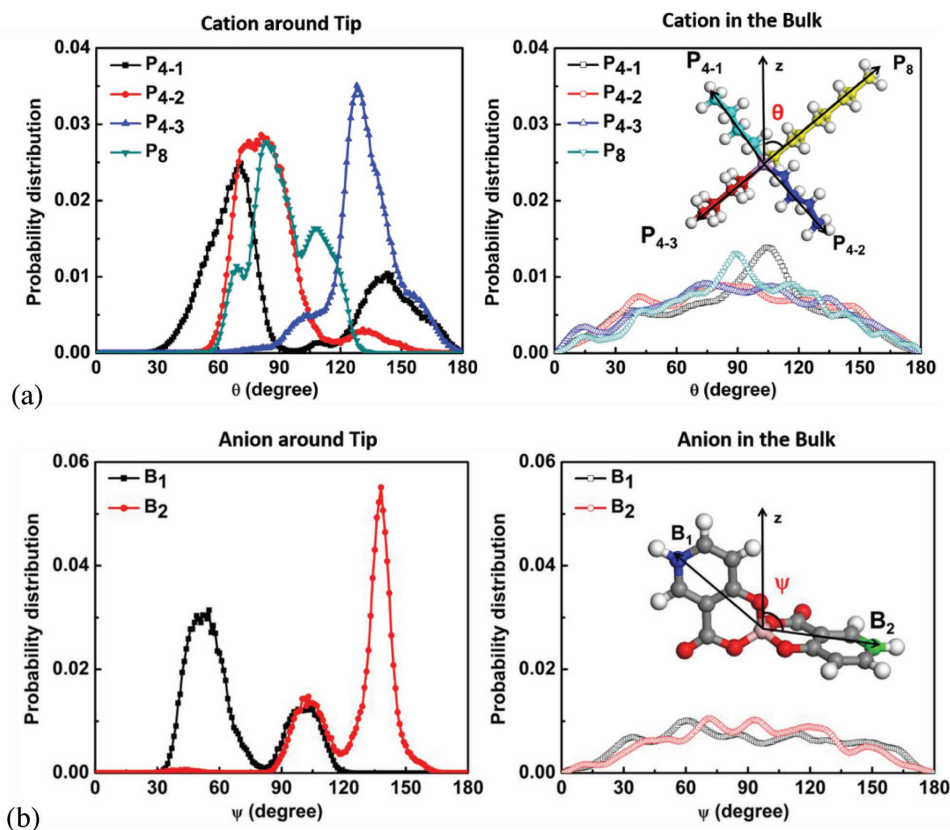


Figure 7. Orientational probabilities of a) the angle θ between the interface normal vector z and the alkyl chain vector P_{4-1} , P_{4-2} , P_{4-3} , P_8 in the cation $[P_{4,4,4,8}]^+$ (left) around the tip and (right) in the bulk system, b) the angle ψ between the interface normal vector z and the vector B_1 , B_2 in the anion $[BScB]^-$ (left) around the tip and (right) in the bulk system. The insets in right panels of (a) and (b) show the vector definition for the orientation analysis of cation and anion, respectively.

between an AFM gold tip and mica.^[29] We highlight a substantially enhanced diffusion coefficient in more strongly confined system, i.e., 5 nm slit width. The diffusion in our cases speeds up in the more strongly confined 5 nm slit pore. As shown in **Table 1**, three slit-pore width 5, 7, and 9 nm were modeled. The diffusion coefficients in 5, 7, and 9 nm slit pore are, respectively, $(3.42\text{--}3.94) \times 10^{-12}$, $(1.19\text{--}1.38) \times 10^{-12}$, and $(1.02\text{--}1.20) \times 10^{-12} \text{ m}^2 \text{ s}^{-1}$. One can safely assume that further increase of the slit-pore width in our systems will produce similar diffusion coefficients to the bulk. The repulsive normal loads at 5, 7, and 9 nm slit-pore walls are, respectively, 707, 788, and 810 nN, indicating that larger normal loads limit the ionic mobility to result in smaller diffusion coefficient. Here it is noteworthy that the loads in calculating diffusion coefficient were produced by varying slit-pore widths, different from that in friction calculations which were applied directly by external normal loads.

Both the acceleration and reduction of the diffusion could occur in more strongly confined systems, but the interpretation of diffusion trends varies. For example, some theorists attributed the diffusion enhancement to the tight hydrogen-bonding network in the liquid molecules and the confined space which shields these hydrogen bonds from fluctuations.^[30] The frictionless surface of the confined walls was also a possible explanation.^[31] But it was unclear whether the flow

enhancement in confined spaces can be attributed simply to the presence of a nearly frictionless surface.^[32,33] The diffusivity increase was also found to be dependent on the liquid polarity.^[34] However, it is uncommon to observe an increased internal energy in the confined liquids, while the self-diffusion in confined conditions exceeds that in the bulk phase. Because stronger intermolecular interactions usually imply smaller diffusivity.^[35] Some investigation ascribed the reduced diffusion in nonpolar fluids to the solidification of confined fluid molecules.^[29] NMR diffusometry experiments showed a higher diffusion coefficient of ethylammonium nitrate ionic liquid confined between parallel glass plates of 4.1 μm as compared with the diffusivity in the bulk. The authors explained the diffusion differences were probably related to electrostatic interactions, hydrogen bonding, nanoscale ordering, structural parameters in confinement.^[36] Another experimental technique, transmission electron microscopy, suggested that the confined fluid mobility was greatly retarded because of the defects of confined walls.^[37]

Therefore, it is difficult to pinpoint a single dominant reason for the diffusion acceleration or reduction, and further experimental and theoretical studies are needed to fully understand varying diffusion phenomena. As for our confined ILs, it is difficult to accurately explain the diffusion acceleration in the confined 5 nm slit pore, but we plan next to calculate the

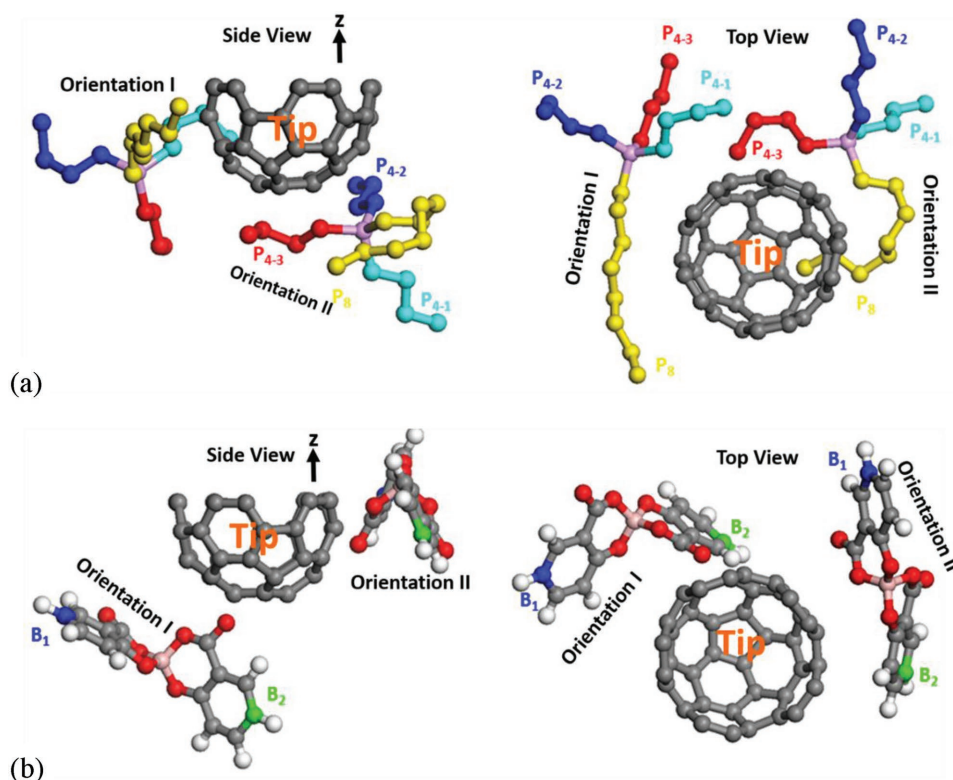


Figure 8. a,b) The sketch map showing the main orientations of P₄₋₁, P₄₋₂, P₄₋₃, P₈ in the cation $[P_{4,4,4,8}]^+$, and B₁, B₂ in the anion $[BScB]^-$, respectively. Here hydrogen atoms were not shown in (a) and (b) to make the illustration more clear.

interactive energy of ILs with the tip and substrate. This can be utilized to evaluate the binding strength of the confined ILs to tell if the surface attractive interactions retard the liquid mobility or not.

Table 1. Normal loads and diffusion coefficients of $[P_{4,4,4,8}][BScB]$ confined in the slit pore (between tip and substrate), calculated by MD simulations.

Slit width [nm]	Normal load ^{a)} [nN]	Diffusion coefficient (D) [$10^{-12} \text{ m}^2 \text{ s}^{-1}$]
9 nm	810	1.02–1.20
7 nm	788	1.19–1.38
5 nm	707	3.42–3.94

^{a)} Normal load in calculating diffusion coefficient was produced by varying slit-pore widths, different from that in friction calculations which were applied directly by external normal loads.

The enhanced diffusion in the smaller width between the tip and the surface in our case led to an increase in the near-surface mobility of ions. The faster overall dynamics of the confined $[P_{4,4,4,8}][BScB]$ system makes the structure change in cations of ILs layer at higher loads. The resulting parallel orientation of ILs is thus responsible for the lower friction force^[38,39] at higher loads than that at lower loads. In addition, the liquid mass transport and diffusion has been found strongly related to the interfacial friction,^[40,41] so that the faster diffusion probably also makes a direct contribution to the negative friction–load dependence in our systems.

3. Conclusions

We have measured negative “friction–load dependence” of IL/oil mixtures at Ti interfaces which is associated with the locally structural changes in ILs during approach of an AFM tip. The negative phenomenon was also confirmed by our NEMD simulations, in which the friction force declines as the normal load increases. Both AFM measurements and NEMD simulations showed larger friction forces than the normal loads, which is attributed to the high viscosity. Neither chemical reactions nor morphologies were observed at IL/oil mixtures at Ti interfaces before and after AFM friction measurements. NEMD simulations revealed a structural reorientation of the studied IL as the normal load increases, i.e., the cation alkyl chains of ILs change the orientation to preferentially stay parallel to the tip

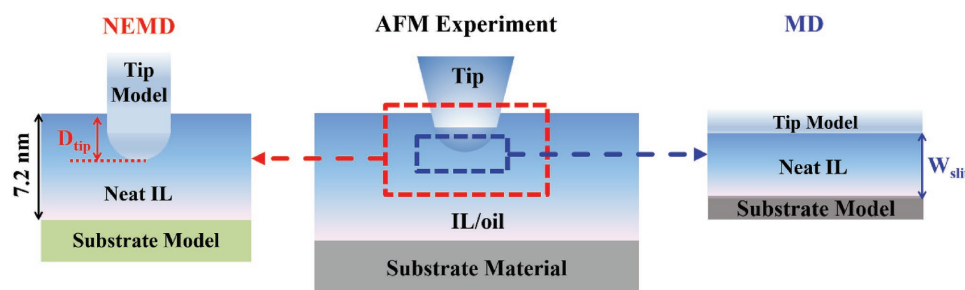


Figure 9. Illustration and connection map of AFM experiments and theoretical simulations.

scanning path, similar to the “blooming lotus leaf.” This reoriented IL structures produce a new sliding interface and reduce the friction force. We further varied the slit-pore width between the tip and the surface using MD simulations to mimic the tip approaching the surface at different loads. And a faster diffusion of ILs was observed in the smaller slit-pore model, which facilitates such structural reorientation at higher normal loads. The resulted new sliding surface to the tip is responsible for the observed smaller friction forces at higher normal loads, i.e., the negative friction–load dependence.

A systematic study would be deserved to be continued, including the type of ionic liquid and base oil; the ratio of ionic liquid/base oil; the hydrophobic/hydrophilic nature of the tip; the sliding speed of the tip as well as the shape and the contacting surface area; the thickness of the lubricant layer (the ionic liquid/base oil mixture); the effect of the supporting substrate.

4. Experimental Section

Samples Preparation: Four ionic liquids were synthesized using methods as reported previously,^[18,42,43] namely, tributylphosphonium bis(salicylate)borate, $[P_{4,4,4,8}][BScB]$; trihexyltetradecylphosphonium bis(salicylate)borate, $[P_{6,6,6,14}][BScB]$; trihexyltetradecylphosphonium bis(mandelate)borate, $[P_{6,6,6,14}][BMB]$; trihexyltetradecylphosphonium bis(oxalate)borate, $[P_{6,6,6,14}][BOB]$. The behavior of the commercial IL trihexyltetradecylphosphonium dicyanamide ($[P_{6,6,6,14}][DCA]$) purchased from Cytec and used as received was also compared. The experimental samples were prepared by mixing 75 wt% ionic liquids in the base oil of diethylene glycol dibutyl ether (DEGDBe, ACROS organics, purity $\geq 99\%$). The IL/oil lubricating film was deposited onto Ti substrate (Sigma-Aldrich, Titanium, foil, thickness 0.25 mm, 99.7% trace metals basis, Product No. 267503) by the spin-coating method at the speed of 5000 rpm for 60 s. It is important to note that the Ti surfaces used here are α -Ti. Based on the X-ray diffraction patterns shown in Figure S1 of the Supporting Information, the α -Ti phase has an hcp crystal structure with lattice parameters of $a = 2.95 \text{ \AA}$, $b = 2.95 \text{ \AA}$, $c = 4.68 \text{ \AA}$, and structure angles of $\alpha = 90^\circ$, $\beta = 90^\circ$, $\gamma = 120^\circ$ (reference code: 00-044-1294).

Friction Measurements: Friction force measurements ($5 \mu\text{m} \times 5 \mu\text{m}$) were performed on a Multimode 8 atomic force microscopy (AFM, Bruker) using Si_3N_4 cantilever tips (DNP-10, A tip, nominal tip radius = 20 nm, tip height = 6 μm) with a nominal spring constant of 0.35 N m^{-1} in contact mode at ambient conditions, and the scan rate was 2 Hz. The normal spring constant of each AFM tip was calibrated using normal deflection sensitivity of the supported cantilever to transform the normal load signals (in V) into true normal load (in N). The calibration procedure was performed whenever a different cantilever was used. The average lateral force, given in terms of an output voltage signal (half-difference of the average lateral deflection signal on the

photodetector of the forward and reverse traces, in V), was transformed into the friction force (in N) from the torsion of the cantilever.^[44] Friction force data presented in this work are the average of five measurements at multiple sample positions.

XPS: XPS (Thermal Scientific, USA) analysis was performed on samples before and after AFM friction measurements to detect possible changes in interfacial compositions due to AFM tip sliding. The analysis was conducted on an Escalab 250Xi Imaging X-ray photoelectron spectrometer with a monochromatic Al K α X-ray small spot source (1486.6 eV).

Molecular Simulations: As illustrated in Figure 9, two types of molecular dynamics calculations were performed to interpret AFM measurements and reveal the molecular level fundamental mechanisms: (a) NEMD of the friction process, and (b) molecular simulation (MD) of the self-diffusion in the confined IL.

Recent advances in NEMD simulations of lubricants and additives have been reviewed by Ewen et al.^[45] Another review of atomic frictions by Dong et al.,^[46] where important parameters, such as materials, surfaces, compliance, contact area, normal load, temperature, and velocity, have been described in the context of both experiments and simulations. In the NEMD calculations, the AFM tip was simplified as a capped single-walled carbon nanotube (SWCNT) with the cap radius of 0.5 nm, to help us understand the underlying mechanisms of the friction behavior, as what had been done in previous studies.^[47,48]

The neat IL $[P_{4,4,4,8}][BScB]$ instead of the IL/oil mixture was calculated. This is because the negative friction–load dependence was also observed in AFM experiments where the neat IL was used, as discussed in Section 2. The simulation box has the dimension of $4.61 \times 4.61 \times 16.5 \text{ nm}^3$, with 150 $[P_{4,4,4,8}][BScB]$ molecules. The bottom of the simulation box was modeled via a fixed Si surface. The NEMD friction simulation protocol started placing the capped SWCNT tip 2.0 nm (the initial insertion depth, D_{tip}) into the IL/Si system to ensure the tip surrounded by the IL molecules, i.e., 5.2 nm away from the IL/Si interface (IL thickness of $\approx 7.2 \text{ nm}$). Varying external normal loads were applied to approach the tip to the IL/Si interface and make intimate contact between the tip and the ILs. And the IL/Si(111) interface was simultaneously slid laterally with the tip at a constant velocity of 1.0 m s^{-1} . The friction force comes from the nonbonded intermolecular interactions, that is, a summation of van der Waals (dispersion) forces and electrostatic interactions. In the NEMD calculation, the capped-SWCNT tip was composed of neutral particles. So, the nonbonded interactions between the IL and the tip are simply dispersion forces. During the calculation, the friction force is calculated by the summation of the total dispersion force of the tip end (colored red in Figure S10, Supporting Information), parallel to the x–y plane, opposite to the tip moving direction. It is noting that the friction calculation started placing the SWCNT tip 5.2 nm away from the IL/Si interface which is much larger than the 1.5 nm cutoff distance, indicative of no interactions between the tip and the Si substrate. The Si surface is therefore simply served herein as a rigid wall to hold simulation box.

About the MD simulation for the self-diffusion of the confined IL, a slit-pore model was applied to calculate self-diffusion properties of confined $[P_{4,4,4,8}][BScB]$, to understand the dynamics of those near-tip IL molecules when the tip approaches the substrate, as demonstrated in

Figure 9. The slit pore was composed of the rigid graphene tip and the rutile (110) substrate with the thickness of 1.17 nm.

Here in diffusion calculations, the surface of the substrate is critical. In order to evaluate the mobility of ILs during AFM tip approaching, the entire ILs diffusion confined between the tip and the substrate has been to consider. This is different from that in NEMD friction calculations, which focused on the tip sliding interface, and the IL molecules around the sliding tip is critical in NEMD simulations. The rutile was thus adopted to model the substrate in MD simulations to mimic the material used in AFM experiments, because native dense titanium dioxide layers^[49] are usually formed on the Ti surface. Three slit widths were investigated: $W_{\text{slit}} = 5, 7, \text{ and } 9 \text{ nm}$. The diffusion coefficient is determined from the Einstein equation: $\langle |r(t) - r(0)|^2 \rangle = 2Dt$, where $r(t)$ is the position of the center of mass of the IL $[P_{4,4,4,8}][\text{BSCB}]$ molecule at time t , D is the diffusion coefficient.^[50]

More simulation details are available in the Supporting Information.

Contact Angle, Density, and Viscosity Measurements: Contact angles were measured using a Fibro 1121/1122 DAT-Dynamic absorption and contact angle tester equipped with a CCD camera at room temperature and ambient humidity. Four drops of IL/oil mixtures were placed on clean titanium surfaces using a manual syringe fixture. An Anton-Paar DMA 4100 density meter was used to measure the density at 20 °C. Viscosity was measured with a Lovis 2000 ME Microviscometer (Anton-Paar falling ball type viscometer) at 20 °C using a sealed sample tube.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

atomic force microscopy, friction, ionic liquid, negative, structural change

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